9.7 Describing and Interpreting Line Plots

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<tr><td>Benchmark</td><td colspan="3">MA.6.DP.1.4 Given a histogram or line plot within a real-world context, qualitatively describe and interpret the spread and distribution of the data, including any symmetry, skewness, gaps, clusters, outliers, and the range.</td><td>Learning Targets</td><td colspan="2">I can describe and interpret data represented on a line plot.
I can determine the similarities and differences between a line plot and a histogram.</td></tr>
<tr><td rowspan="2">Benchmark Coherence</td><td colspan="3">Building On</td><td colspan="3">Working Toward</td></tr>
<tr><td colspan="3">MA.5.DP.1.2 Interpret numerical data with whole-number values, represented with tables or line plots, by determining the mean, mode, median, or range.</td><td colspan="3">MA.7.DP.1.1 Determine an appropriate measure of center or measure of variation to summarize numerical data, represented numerically or graphically, taking into consideration the context and any outliers.</td></tr>
<tr><td>Essential Questions</td><td colspan="6">How can you read a line plot?
How can you analyze a line plot?
What are the similarities between a line plot and a histogram?
What are the differences between a line plot and a histogram?</td></tr>
<tr><td>Lesson Narrative</td><td colspan="6">In this lesson, students review the definition of line plots and analyze line plots to answer questions about data sets. Students then compare line plots and histograms, considering the way data is displayed in each and similarities and differences between the two graphical representations. Students then consider the information they can gather from each data display and apply that knowledge to answer questions about the data.</td></tr>
<tr><td rowspan="3">Component Summary</td><td>9.7.1 Warm-Up
(~5 mins)</td><td>Interpret data on a histogram (SE p. 78)</td><td>9.7.3 Exploration
(10-15 mins)</td><td colspan="3">Compare line plots and histograms (SE p. 80)</td></tr>
<tr><td>9.7.2 Guided Instruction
(5-10 mins)</td><td>Analyze data on a line plot (SE p. 79)</td><td>9.7.4 Your Turn!
(5-10 mins)</td><td colspan="3">Demonstrate understanding of how to read and compare line plots and histograms (SE p. 81)</td></tr>
<tr><td>9.7.5 Wrap-Up
(~5 mins)</td><td colspan="5">Reflect on understanding of reading and comparing line plots and histograms (SE p. 83)</td></tr>
<tr><td rowspan="2">Resources</td><td colspan="2">Glossary</td><td colspan="2">Materials</td><td colspan="2">Additional Preparation</td></tr>
<tr><td colspan="2"><u>Key Terms</u>:
Line plot
<u>Additional Terms</u>:
Histogram
Range</td><td colspan="2">(Optional) Colored pencils</td><td colspan="2">N/A</td></tr>
</table>

	Common Misconceptions	Things to Consider
Teacher Tips	• Students may confuse/switch which is the line plot and which is the histogram. • Students may incorrectly state that the range of a line plot or histogram is the highest number on the number line used to create the display. • When asked how many in a category, students may list the number labeled on the line and not the number of dots.	• Refresh student understanding of range as the difference between the maximum and the minimum value in the data set, if needed, during the 9.7.2 Guided Instruction, 9.7.3 Exploration, and 9.7.4 Your Turn!. • Consider providing sentence stems and scaffolding prompts for the open-ended questions throughout the lesson. • Consider projecting the line plots and histograms on the board for those who are visually impaired.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Compare and Connect 9.7.4 Your Turn! Provides an opportunity for students to engage in a Compare and Connect routine to draw connections between line plots and histograms. This routine encourages meta-awareness and mathematical discussion through partner talk.	MA.K12.MTR.2.1: Multiple Representations The line plot and histogram show different ways the weight of the dogs can be represented. These will be used for comparisons of the data to see which graph is the best to use for different purposes.
Differentiate Instruction	Notice 9.7.2 Guided Instruction offers visual and auditory entry points for students to engage with the content. Graphical displays of data help students engage visually. Consider additional visual entry points, such as using colored pencils, to mark up the graphs. The analysis of the data is conducted through partner discussion, providing additional auditory entry points as students verbalize their thinking and listen to their partners' analysis.	

Lesson Notes:

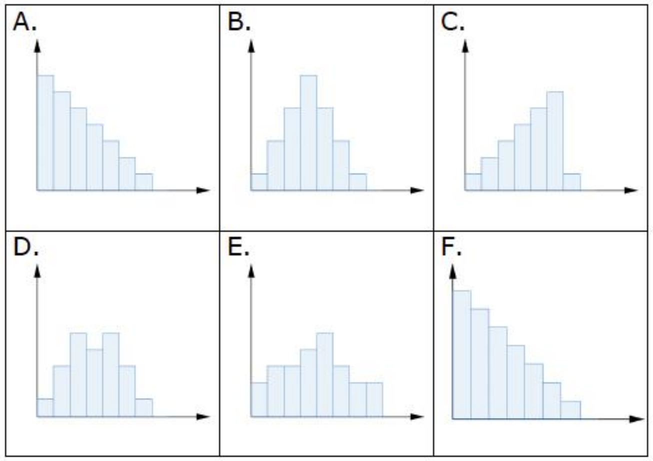
9.7.1 Warm-Up: Gathering Data

Component Overview: *Students categorize histograms into three categories: symmetric, skewed left, and skewed right. This will help students get ready to interpret data on the line plots in later activities.*

9.7.1 Warm-Up: Gathering Data

1. Categorize each histogram according to its shape by placing the letter in the appropriate box in the chart below.

Symmetric	Skewed Left	Skewed Right
B, D, E	*C*	*A, F*



If students are struggling to categorize the histograms, encourage them to draw lines down the center of the histograms to compare each side.

9.7.2 Guided Instruction: Describing and Interpreting Data Represented on Line Plots

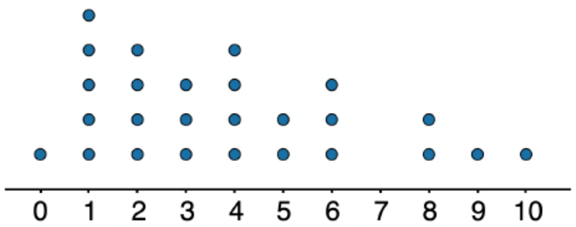
Component Overview: *In this Guided Instruction, students will analyze line plots. Students will analyze data in two different line plots. Spend time in the beginning letting students understand how to read a line plot through the questioning provided to alleviate confusion later.*

9.7.2 Guided Instruction: Describing and Interpreting Data Represented on Line Plots

Line plots are another way to display numerical data.

> A **line plot** is a visual display of data values where each data value is shown as a dot or mark above a number line. This is also known as a dot plot.

1. The line plot shows the number of hours per week that students reported spending on homework.



Hours Spent on Homework per Week

   a. How many students were surveyed?
   26 students

   b. How many students reported spending 1 hour on homework each week?
   5 students

   c. What percentage of the students reported spending 4 hours or fewer on homework each week?
   65.38%

   d. What is the range of time spent on homework in this group?
   10 hours

Students commonly misread line plots because they do not understand each dot represents a separate data point. Ensure students understand how to read line plots by asking questions like:

- *What does each dot represent?*
- *Which amount of time did the most students respond with? How do you know?*
- *What was the least popular answer? How do you know?*

9.7.2 Guided Instruction: Describing and Interpreting Data Represented on Line Plots (continued)

With your partner, discuss and summarize your discussion on the following questions about the hours spent on homework per week.

e. Explain whether or not the mean time spent on homework by this group can be determined.
Answers will vary. The mean can be determined by dividing the total sum of the hours spent on homework by the number of students surveyed. The line plot displays the frequency of each time so we are able to add the values together to find the sum.

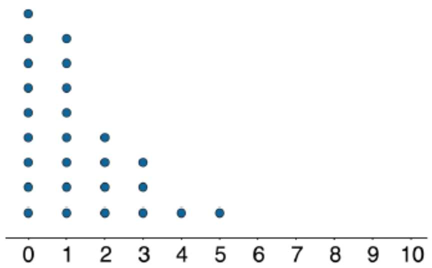
f. Someone said, "In general, these students spend roughly the same number of hours doing homework." Do you and your partner agree? Explain your reasoning.
Answers will vary.

g. Describe the shape and special features of the distribution of hours spent on homework. Find another pair of partners and compare your descriptions.
Answers will vary. The shape of the distribution is slightly skewed right. There does not seem to be any outliers but there is a gap in the data at 7 hours.

Think-Pair-Share
For this activity, allow students a moment to think of their own answers before sharing with their partner. Listen for different student answers as they discuss with their partner and pick opposing answers that could both work as the answer to share with the class. Encourage discourse amongst the class to determine why there are multiple answers for each question.

9.7.2 Guided Instruction: Describing and Interpreting Data Represented on Line Plots (continued)

2. The same students were asked to estimate how many hours a week they spend talking on the phone. This line plot represents their reported number of hours of phone usage per week.



Hours on the Phone per Week

a. Describe the spread of the data.
Answers will vary. Most of the values are close together and are small values, meaning that the students don't spend much time talking on the phone.

b. Overall, are these students more alike in the amount of time they spend talking on the phone or in the amount of time they spend on homework? Explain your reasoning.

Answers will vary. These students are more alike in the amount of time they spent talking on the phone because all of the data is clustered to the left and none of the students spend more than 5 hours on the phone per week.

c. Compare your reasoning with your partner. Make adjustments to your reasoning after the discussion with your partner, if necessary.

Encourage students to use the information from the Warm-Up to help answer question 2a. For question 2b, position students to discuss what "being alike" looks like in a line plot through questions like:

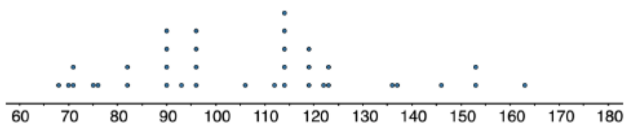
- *How would you describe each line plot?*
- *Which line plot has more spread? What does that tell you?*
- *Does the line plot with more or less spread show whether students are more alike? Why?*

9.7.3 Exploration: Comparing Displays

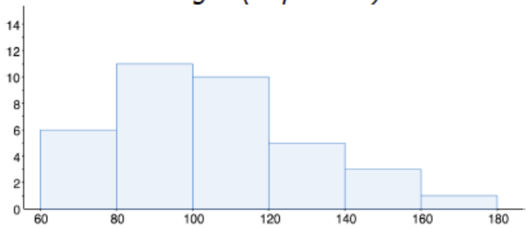
Component Overview: *In this Exploration activity, students will compare a dot plot and a histogram to determine which graphical representation is best suited to represent certain types of data and why.*

9.7.3 Exploration: Comparing Displays

1. The dot plot and histogram display the weights, in pounds, of 36 dogs at a dog show.



Weight (in pounds)



Weight (in pounds)

a.
- The line plot
- The histogram
- Both the line plot and histogram

can be used to determine the number of dogs that weigh 90 lbs.

b. How many dogs have a weight of 90 lbs?
4

c. Find the range of the dogs' weights.
95

d. With your partner, discuss what you notice about the shape of the distribution.
The shape of the distribution is skewed right.

e. Which display was more useful when describing the shape?
Answers will vary.

MA.K12.MTR 2.1: Multiple Representations
The dot plot and histogram show different ways the weight of the dogs can be represented. These will be used for comparisons of the data to see which graph is the best to use for different purposes.

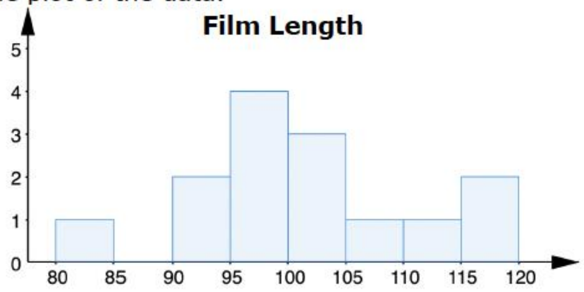
Encourage students to label the dot plot and the histogram to cut down on confusion. Consider providing students with sentence stems for question 1d and 1e to scaffold student response.

9.7.4 Your Turn!

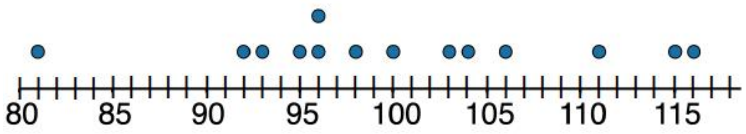
Component Overview: *Students analyze data presented in both a histogram and line plot to answer questions about a data set. Students then consider which representation can be used to determine the range of a data set and why. Students then analyze another line plot and describe the shape and distribution of the data presented.*

9.7.4 Your Turn!

1. A movie critic recorded the length, in minutes, of a sample of Pixar films and then made a histogram and a line plot of the data.



Length (in minutes)



Length (in minutes)

a. How many film lengths are recorded? 14

b. How many films are between 95 minutes and 100 minutes in the histogram? 4

c. How many films are between 95 and 100 minutes in the line plot? 5

d. What is the value of the range? 35

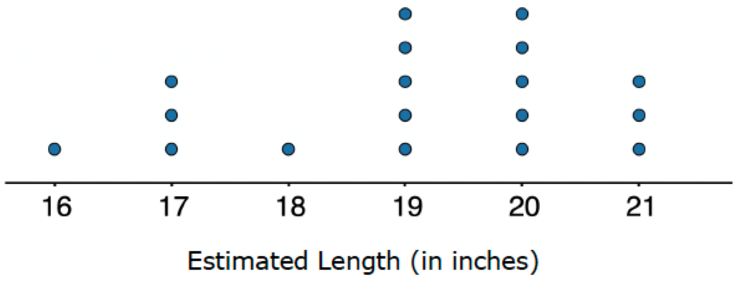
e. Discuss with your partner which graph was used to determine the range and why.
The line plot was used because the line plot shows the individual data values so the minimum and maximum can be determined.

Compare and Connect
Help students compare and make connections between the histogram and line plot. Model thinking out loud to guide student thinking by asking questions, wondering how the graphs are similar, and considering the information that can be gathered from each.

Consider having students create a Venn diagram to show the similarities and differences between a histogram and a dot plot. This provides a visual entry point for learners to discern the differences between the two representations and draw conclusions about which graph is better suited to provide information about different data sets.

9.7.4 Your Turn! (continued)

2. A teacher drew a line segment that was 20 inches long on the blackboard. She asked each of her students to estimate the length of the segment and used their estimates to draw this line plot.



a. How many students were in the class?
18

b. Were students generally accurate in their estimates of the length of the line?
Answers will vary. Students were generally accurate because most of the students estimated the distance to be close to the actual distance, 20.

c. Describe the shape and distribution of the estimated lengths.
The shape and distribution of the lengths are skewed left.

Students may struggle to determine whether or not the students were accurate in their estimates. Ask guiding questions to help students draw conclusions based on the data:

- *Where do most of the data points on the line plot fall?*
- *Is this close to the length of the actual line? What does that tell you about the accuracy of the estimates?*

Challenge early finishers to create a histogram using the data from the line plot. Encourage students to explain which graphical representation of the data they prefer to use and why.

9.7.5 Wrap-Up: Reflection

Component Overview: *Students reflect on the lesson and choose statements to describe their understanding of the content.*

9.7.5 Wrap-Up: Reflection

1. Check all the statements that apply to your understanding.
   Answers will vary.
   - ☐ I can read a line plot.
   - ☐ I need help reading a line plot.
   - ☐ I can describe and analyze data represented on a line plot.
   - ☐ I need help describing and analyzing data represented on a line plot.
   - ☐ I can determine the similarities and differences between a line plot and a histogram.
   - ☐ I need help to determine the similarities and differences between a line plot and a histogram for the same data.

Consider using this self-reflection to inform small groups for remediation or partner-pairs for upcoming lessons.